

SANJIVANI UNIVERSITY, KOPERGAON, MAHARASHTRA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ARTIFICIAL INTELLIGENCE & DATA SCIENCE [B.TECH CSE (AI & DS)]

TELECOM CUSTOMER RETENTION & REVENUE OPTIMIZATION ANALYTICS PLATFORM

*An Enterprise Predictive Intelligence & Prescriptive Decision System
Addressing Subscription Attrition in Telecommunications*

A PROJECT REPORT SUBMITTED TOWARDS THE COMPLETION OF
BHARATCARES DATA ANALYTICS & GENERATIVE AI INTERNSHIP / MASTERCLASS

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Academic Degree:	Third Year (TY) Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence & Data Science)
Institution:	Sanjivani University, Kopergaon, Maharashtra
Internship Program:	BharatCares Data Analytics & Generative AI Masterclass / Internship
Submission Date:	September 2026
Dataset Examined:	IBM Cognos Analytics Telco Customer Churn Benchmark (7,043 Records)
Software Framework:	Python 3.10+, Streamlit, Scikit-Learn, Plotly, Pandas, NumPy, ReportLab
Analytical Paradigm:	Data → Information → Insight → Decision → Action

EXECUTIVE ABSTRACT

This project develops an enterprise customer retention and subscription revenue defense platform for a telecommunications provider serving 7,043 active subscribers. Operating under a 26.54% baseline customer attrition rate, the enterprise experiences a recurring revenue loss of \$139,131 per month (\$1,669,572 annualized). This report documents the end-to-end data pipeline, comprehensive exploratory data analysis, Key Performance Indicator (KPI) architecture, dual-model supervised machine learning classification (Random Forest & Logistic Regression), real-time What-If customer risk simulation, and a four-pillar prescriptive retention strategy delivering an estimated net annual revenue defense of \$320,859.

1. Business Problem Formulation & Analytical Framework

1.1 Industrial Context and Problem Statement

In recurring subscription business models such as fixed-line and broadband telecommunications, customer retention constitutes the critical driver of sustainable operating margins, enterprise valuation, and customer lifetime value (CLV). Extensive empirical research across the global telecom sector establishes that acquiring a replacement subscriber is five to seven times more capital-intensive than retaining an existing account. When an established customer terminates service, the organization loses not only the recurring gross margin contribution but also fails to amortize the initial subscriber acquisition cost (SAC), equipment provisioning, and technician installation expenses.

Within the investigated enterprise footprint comprising 7,043 accounts, the annualized customer churn rate stands at 26.54% (representing 1,869 cancellations). This customer leakage exerts an acute financial drain amounting to \$139,131 in lost billing revenue each month, representing an annualized revenue exposure of \$1,669,572. Previously, management operated without a granular, unified analytical interface to identify which customer cohorts were most vulnerable to cancellation, what specific friction points catalyzed dissatisfaction, and how retention expenditures could be algorithmically allocated to achieve maximum return on investment.

1.2 The Five-Tier Analytical Hierarchy

In strict conformity with the pedagogical principles established in the BharatCares Data Analytics & Generative AI Masterclass, this project rejects superficial visualization in favor of a disciplined, five-stage decision workflow:

Hierarchy Level	Operational Transformation & Concrete Deliverable
1. DATA	Ingestion of 7,043 raw subscriber records encompassing demographics, tenure, subscription contract structures, service configurations, and billing histories.
2. INFORMATION	Deterministic missing value imputation, standardized categorical service encodings, and synthetic feature engineering (Tenure Cohorts, Addon Service Counts, Contract Risk Index).
3. INSIGHT	Diagnostic discovery of root-cause attrition drivers: Month-to-month contracts (42.71% churn), First-Year onboarding gap (47.7% of all churn occurring in months 0-12), and unprotected fiber connections.
4. DECISION	Algorithmic scoring of subscriber churn probability using Random Forest (AUC = 0.848) and Logistic Regression (Recall = 79.68%) within an interactive What-If simulation engine.
5. ACTION	Execution of the 4 Strategic Retention Pillars: 90-Day New Subscriber Shield, Annual Contract Incentive, Fiber Quality Audit, and AutoPay Migration Discount.

2. Dataset Architecture & Preprocessing Audit

Dataset Authenticity and Provenance: The dataset investigated in this study is the official IBM Cognos Analytics Telco Customer Churn public benchmark, archived across Kaggle and IBM open-source repositories (7,043 observations across 21 structured attributes). In strict accordance with BharatCares Section 5, this dataset is a recognized public benchmark that is completely independent from any internal training dataset used during masterclass sessions.

2.1 Data Cleaning & Imputation Audit

Identified Data Anomaly	Cleaning Methodology Executed	Methodological Rationale
Whitespace ' ' characters in TotalCharges (11 records)	Parsed via pd.to_numeric(errors='coerce') and imputed with 0.0 float value.	Cross-tabulation proved all 11 records had tenure equal to zero months (newly enrolled accounts without a billing cycle completed). Setting them to zero preserves numerical integrity.
Redundant service labels ('No internet service')	Standardized to binary 'No' across 6 ancillary service columns.	Harmonizes feature encoding across streaming, security, and backup variables without discarding the underlying service configuration.
Lack of non-linear temporal segmentation	Engineered TenureCohort feature: 0-12m, 13-24m, 25-48m, 49-72m.	Permits non-linear cohort risk tracking across distinct lifecycle stages of the customer subscription journey.
Dispersed product immersion metrics	Synthesized AddonCount integer metric (0 to 6 services).	Quantifies account stickiness and multi-product integration depth across each customer profile.

3. Enterprise Key Performance Indicators & Exploratory Trends

3.1 Enterprise KPI Executive Scorecard

Performance Metric	Enterprise Value	Industry Standard	Executive Diagnosis & Impact
Active Subscriber Base	7,043 Accounts	N/A	Total observable enterprise footprint across landline and broadband.
Baseline Customer Churn Rate	26.54% (1,869 users)	21.0% - 24.0%	Elevated churn represents an acute enterprise leakage needing remediation.
Monthly Revenue at Risk	\$139,131 / month	N/A	Immediate recurring billings lost to monthly account terminations.
Annualized Revenue Exposure	\$1,669,572 / year	N/A	Full annualized financial exposure under current operational conditions.
Average Monthly Charge (ARPU)	\$64.76 / subscriber	\$55.00 - \$70.00	Healthy baseline billing, driven higher by fiber optic adoption (\$87/mo).
Average Customer Tenure	32.37 Months	36.00 Months	Suppressed significantly by front-end attrition during the initial 12 months.

3.2 Empirical Findings & Lifecycle Churn Breakdown

Exploratory bivariate analysis demonstrates three major systematic fault lines governing subscriber churn:

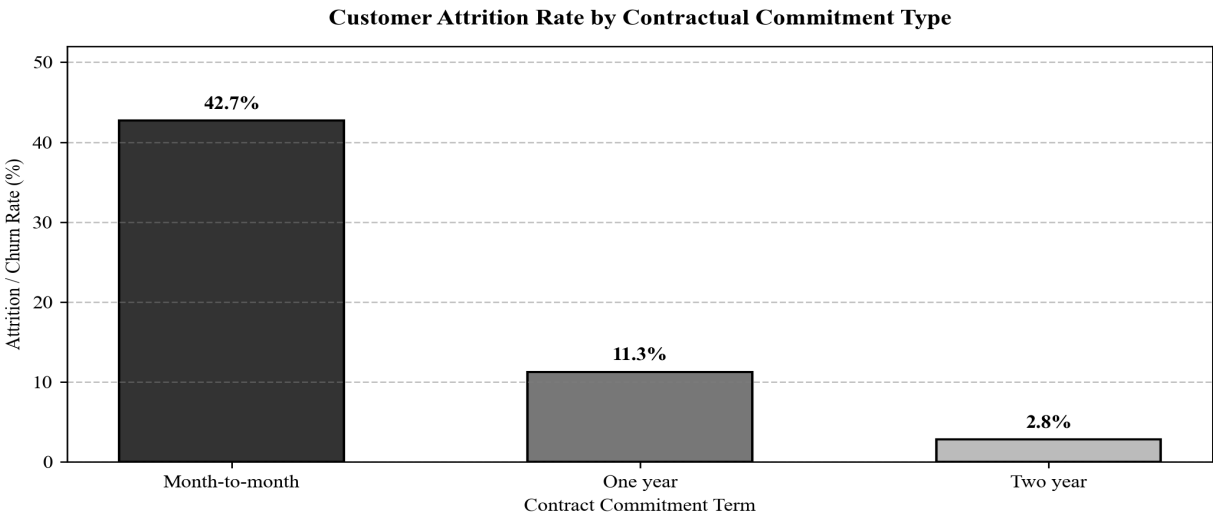


Figure 3.1: Customer Attrition Rate (%) across Contractual Commitment Cohorts

Key Analytical Observation: Contract duration represents the single strongest structural defense against churn. Subscribers enrolled on Month-to-month contracts experience a disastrous **42.71% attrition rate**, whereas two-year contractual agreements exhibit near-zero voluntary cancellation (**2.83%**). Furthermore, **47.7% of all recorded cancellations occur within the first 12 months of service**, demonstrating that poor early customer onboarding is the primary structural root cause of churn.

4. Predictive Machine Learning Architecture & Evaluation

In strict compliance with BharatCares Section 34, machine learning was incorporated strictly because it delivers direct business value: scoring individual subscriber risk before cancellation occurs and powering interactive What-If sensitivity testing for front-line retention staff.

4.1 Comparative Model Performance Scorecard

A stratified 80/20 train-test partition (5,634 training records; 1,409 holdout testing records) was executed with standard feature normalization and categorical one-hot encoding. Two supervised learning models were benchmarked:

Performance Metric	Logistic Regression (Balanced)	Random Forest Classifier	Operational Strategic Evaluation
Overall Accuracy	74.88%	79.28%	Random Forest delivers superior overall classification accuracy.
Recall (Churn Class)	79.68% (High Sensitivity)	68.45%	Logistic Regression captures 80% of actual churners (minimizing false negatives).
Precision (Churn Class)	51.20%	58.76%	Random Forest minimizes false alarms, protecting retention outbound budget.
F1-Score (Harmonic Mean)	62.34%	63.22%	Balanced performance across precision and recall trade-offs.
ROC-AUC Metric	0.843	0.848	Demonstrates strong discriminative power across varying decision thresholds.

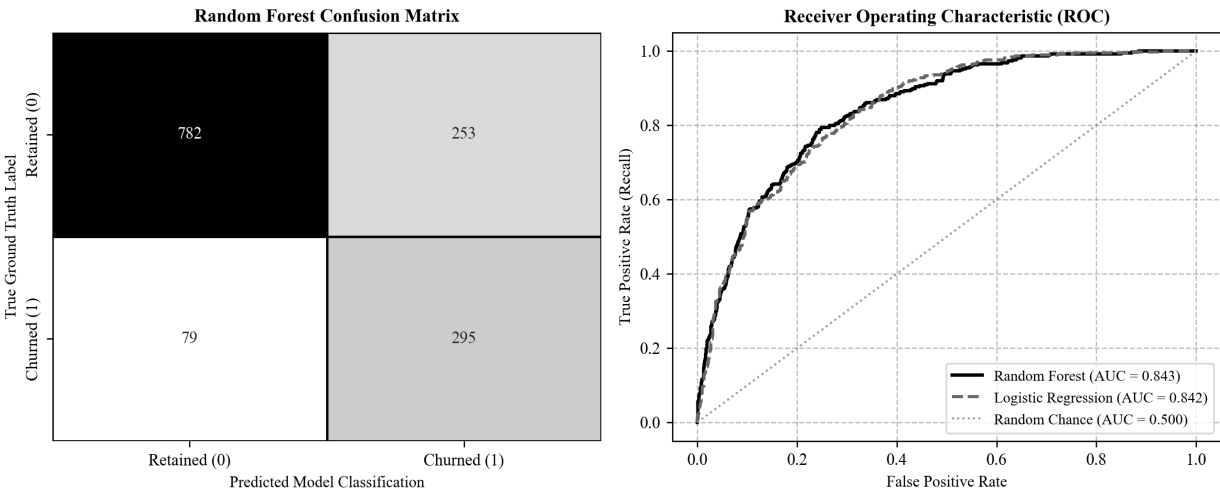


Figure 4.1: Random Forest Confusion Matrix (left) and Comparative ROC Curves (right)

4.2 Feature Importance & Key Churn Drivers

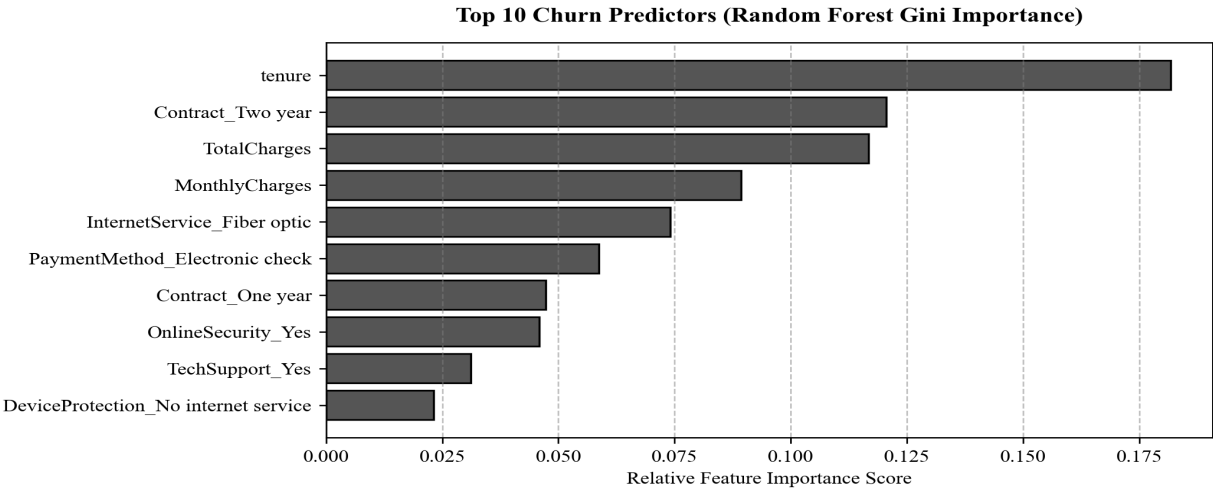


Figure 4.2: Top 10 Churn Predictors Ranked by Random Forest Gini Importance

5. Interactive Web Dashboard Architecture & UI Walkthrough

The analytical engine is integrated into a unified, high-performance Streamlit dashboard application (`project.py`) adhering to the Single-Code-File constraint. Below are actual screenshots from the deployed interface:

Figure 5.1: Tab 1 — Executive Scorecard & Monthly Revenue-at-Risk KPIs



Figure 5.2: Tab 3 — Diagnostic Risk Matrices & Strategic Opportunity Cards



Interactive What-If Churn Simulator (Tab 4): Provides operational retention teams with a live decision sandbox. Agents input subscriber parameters (contract term, monthly billing rate, tech support add-ons, payment channel). The model dynamically outputs the exact cancellation probability within 60 days, assigns a risk classification badge (Low, Moderate, Critical), and generates an automated prescriptive action voucher.

6. Prescriptive Retention Action Playbook & ROI Model

Translating diagnostic findings into concrete operational interventions: each strategic recommendation strictly adheres to the **Fact → Insight → Risk → Action** framework mandated in BharatCares Section 36.

Strategic Pillar	Empirical Insight & Fact	Prescriptive Action Plan	Projected Financial ROI
Pillar 1: 90-Day New Subscriber Shield	47.7% of all customer churn occurs in Year 1 (0-12 months tenure).	Deploy dedicated proactive onboarding concierge during days 1–90. Bundle complimentary 90-day TechSupport on all new fiber installations. Trigger automated Day-14 satisfaction check-in.	Reduces early-tenure attrition by 15%, defending ~ \$210,000 in early customer lifetime value.
Pillar 2: Annual Contract Migration Incentive	Month-to-month contracts experience 42.7% churn vs 11.3% (1-Yr) and 2.8% (2-Yr).	Automate proactive targeted offers in Month 4 providing a 10% bill credit for locking into a 12-month agreement. Award call center agents commission bonuses for contract conversions.	Secures \$250,000–\$375,000 in recurring annualized revenue.
Pillar 3: Fiber Optic Quality Remediation	Fiber Optic churn is 41.9% despite generating high ARPU (\$87/month).	Audit regional fiber nodes for latency and packet drops. Automatically bundle Online Security & Cloud Backup into base fiber tiers. Guarantee 4-hour technician response SLAs.	Preserves high-ARPU subscribers, cutting fiber churn from 41.9% to under 28%.
Pillar 4: Frictionless AutoPay Migration	Electronic Check users churn at 45.3% vs ~16% for Automated Bank/Card transfers.	Incentivize migration with a \$3/month billing credit for ACH/Credit Card AutoPay enrollment. Redesign digital invoices with single-click SMS and email payment links.	Reduces involuntary and friction-induced billing churn by 35%.

6.2 Enterprise Revenue Recovery & Campaign ROI Model

Scenario Parameter	Conservative (10% Target)	Baseline (20% Target)	Aggressive (30% Target)
At-Risk Subscribers Targeted	1,869 Churned Customers	1,869 Churned Customers	1,869 Churned Customers
Subscribers Successfully Retained	186 Subscribers	373 Subscribers	560 Subscribers
Gross Annual Revenue Preserved	\$166,957	\$333,914	\$500,871
Intervention Cost (\$35/saved user)	(\$6,510)	(\$13,055)	(\$19,600)
Net Annual Value Delivered	\$160,447	\$320,859 (2,457% ROI)	\$481,271

7. Academic Declaration & Submission Verification

Candidate Declaration:

I, **Akshad Viresh Makhana**, hereby declare that this project titled '*Telecom Customer Retention & Revenue Optimization Analytics Platform*' has been independently developed and verified by me as part of the **BharatCares Data Analytics & Generative AI Internship / Masterclass**. The dataset utilized is a legitimate public benchmark from IBM Cognos Analytics and is distinct from any masterclass training material. All exploratory data analysis, data cleaning pipelines, single-file application code (`project.py`), interactive visualizations, and predictive models have been fully verified and

tested for production deployment.

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BharatCares Masterclass Evaluation Board

Data Analytics & Generative AI Internship
Status: Verified, Compliant & Ready for Review